

Milestone 3 Report

Group 09

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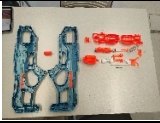
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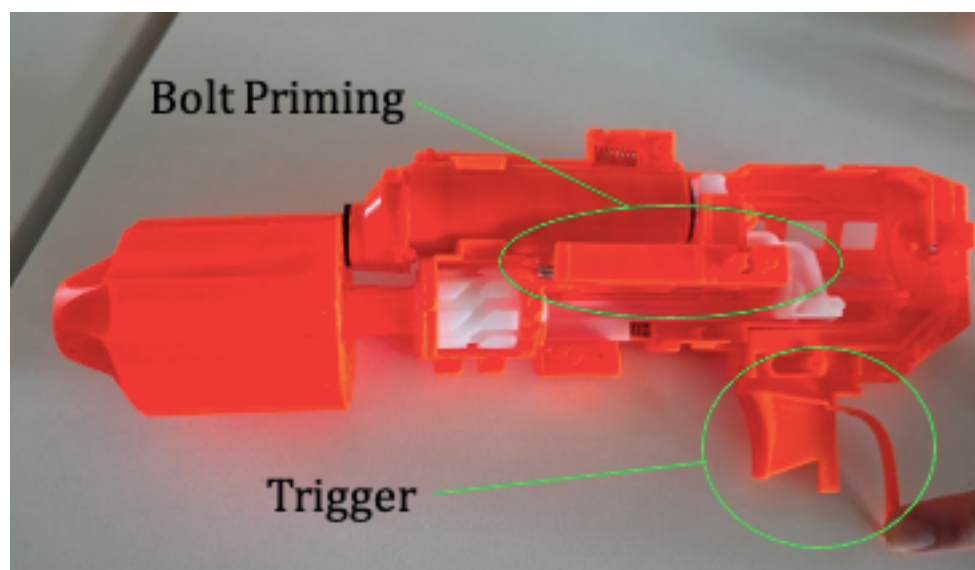
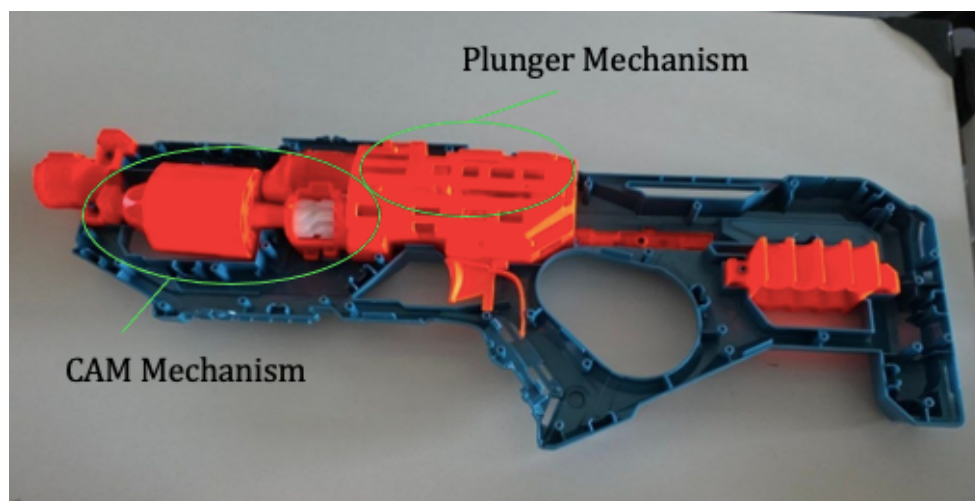
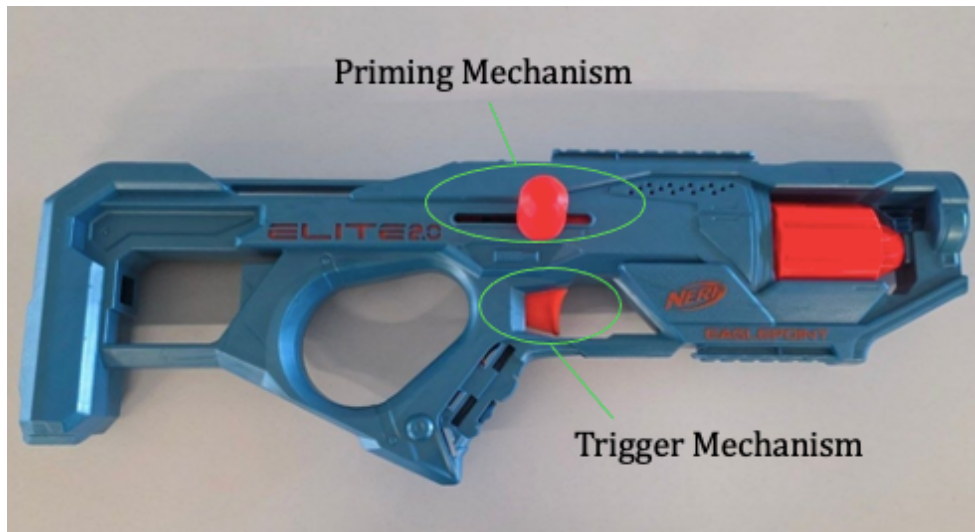
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Device Dissection Procedure

Nerf Gun Disassembly						
Steps	Description	Tools	Pictures			
1.	Inspect for damages or missing parts		Front	Back	Left	Right
						
2	Unscrew the 12 screws in the nerf gun circled in red	PH2 Phillips Screwdriver				
3.	Unscrew the knob on the right side of the nerf gun in an anticlockwise motion					
3.	Remove the frame of nerf gun					
4.	Remove the detachable parts of the nerf gun					

System Overview



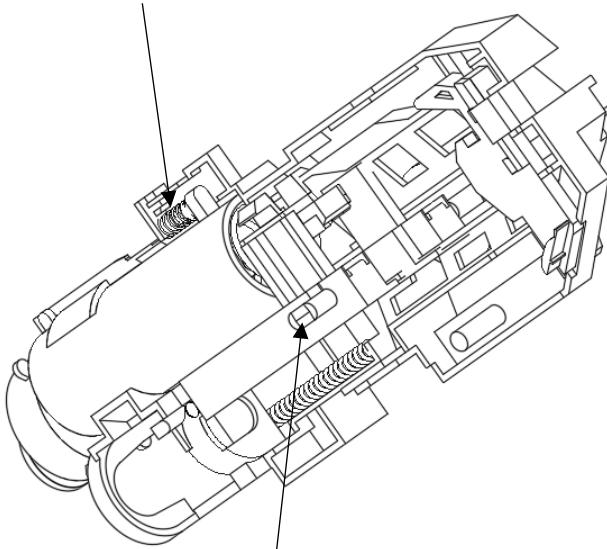
Priming Mechanism Subsystem Overview Diagram

Top Spring (Compressive)

- Attached onto the plunger housing to absorb the shock from the plunger when it fires

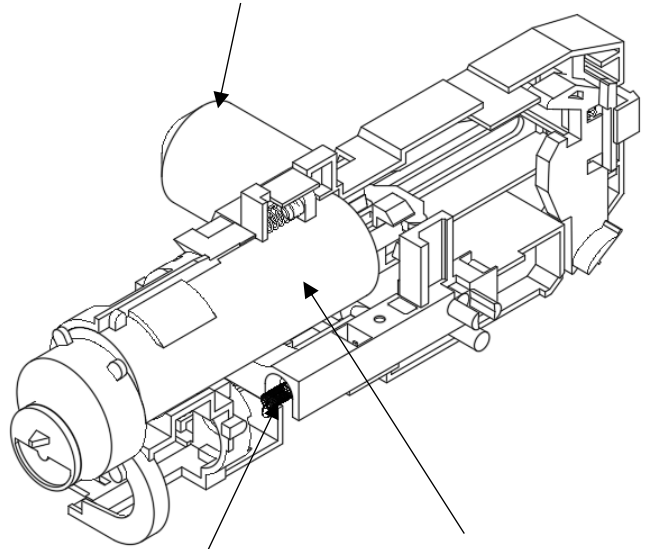
Priming Bolt Handle

- Screwed onto the "handle link" component with a steel threaded rod
- Slides linearly along with length of the frame



Handle Link

- Contains threaded grooves on either ends for the bolt handle to screw in
- Attaches onto the plunger spring housing when priming



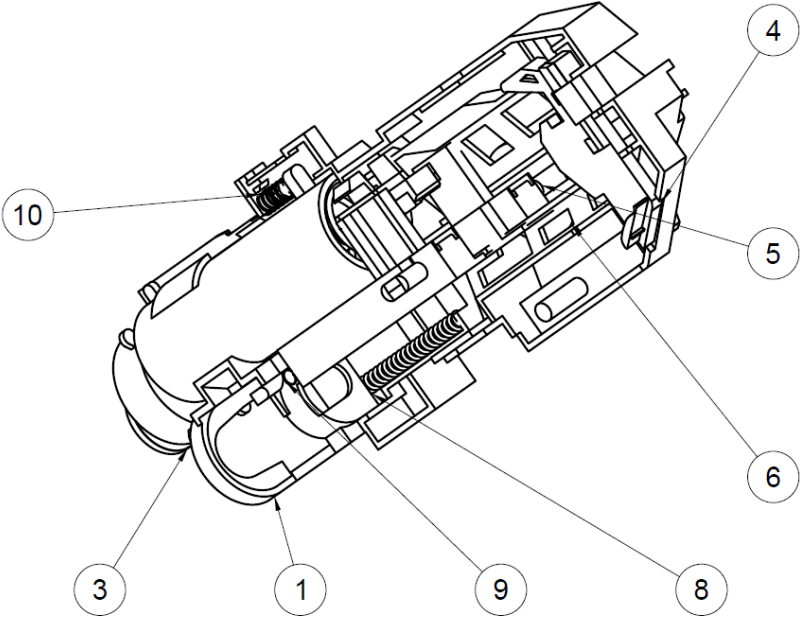
Plunger Housing and Plunger Spring (Compressive)

- Plunger housing houses plunger spring (inside the housing)
- Plunger spring transfers the compressed air onto the dart

Helical Spring (Extension)

- Attached onto the "handle link" component to prevent lateral displacement when priming

BOM Sheet (Priming Mechanism)

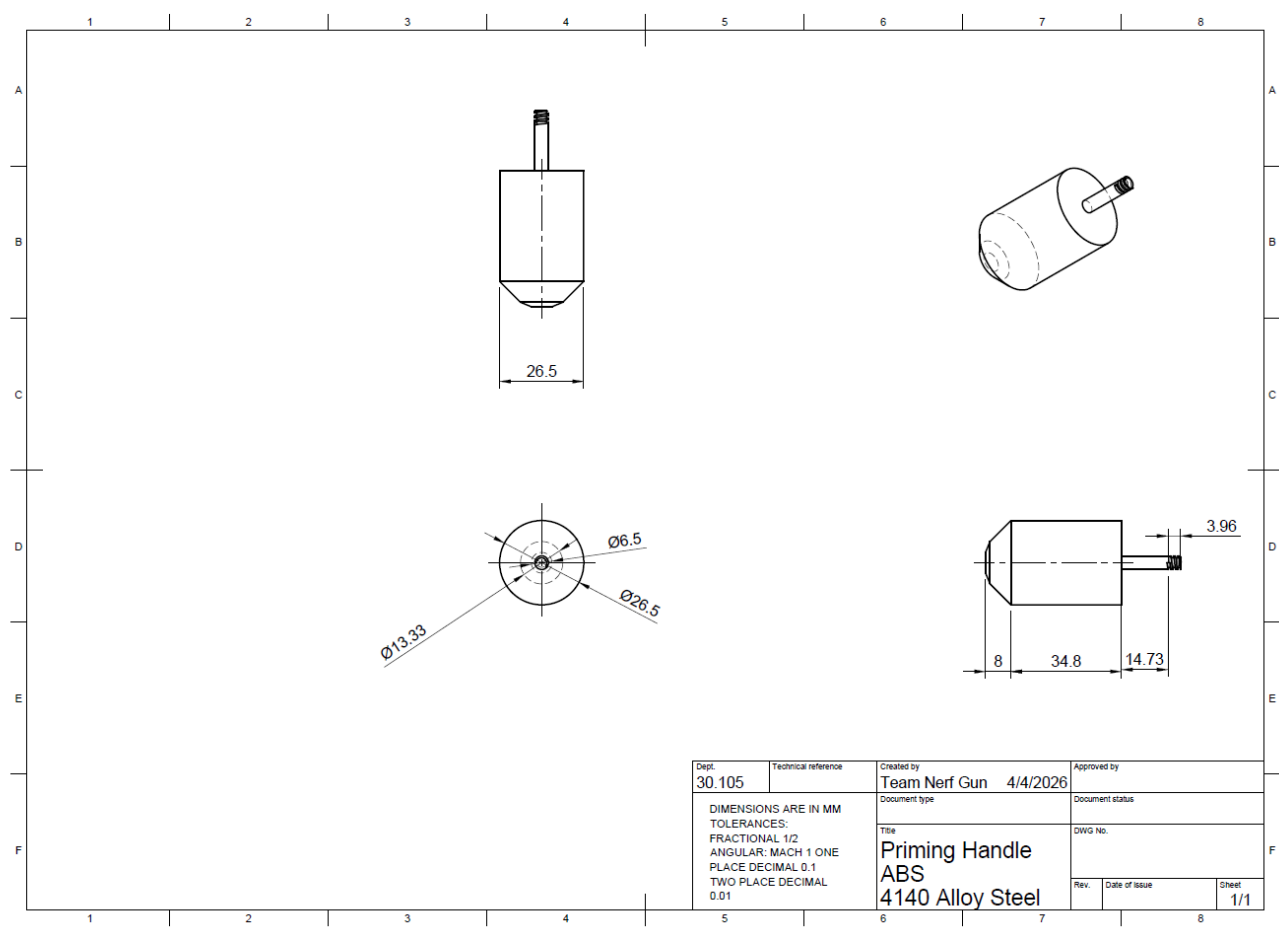


Exploded view diagram of a Priming Mechanism assembly. The diagram shows the internal components and their assembly sequence, with numbered callouts 1 through 10. The components are: 1. Internal Frame, 2. Plunger_gasket_su bassy, 3. Trigger Release, 4. HandleLink, 5. Slide Link, 6. Springs, 7. HelicalSpring, 8. TopSpring, 9. Internal Frame, and 10. Plunger_gasket_su bassy.

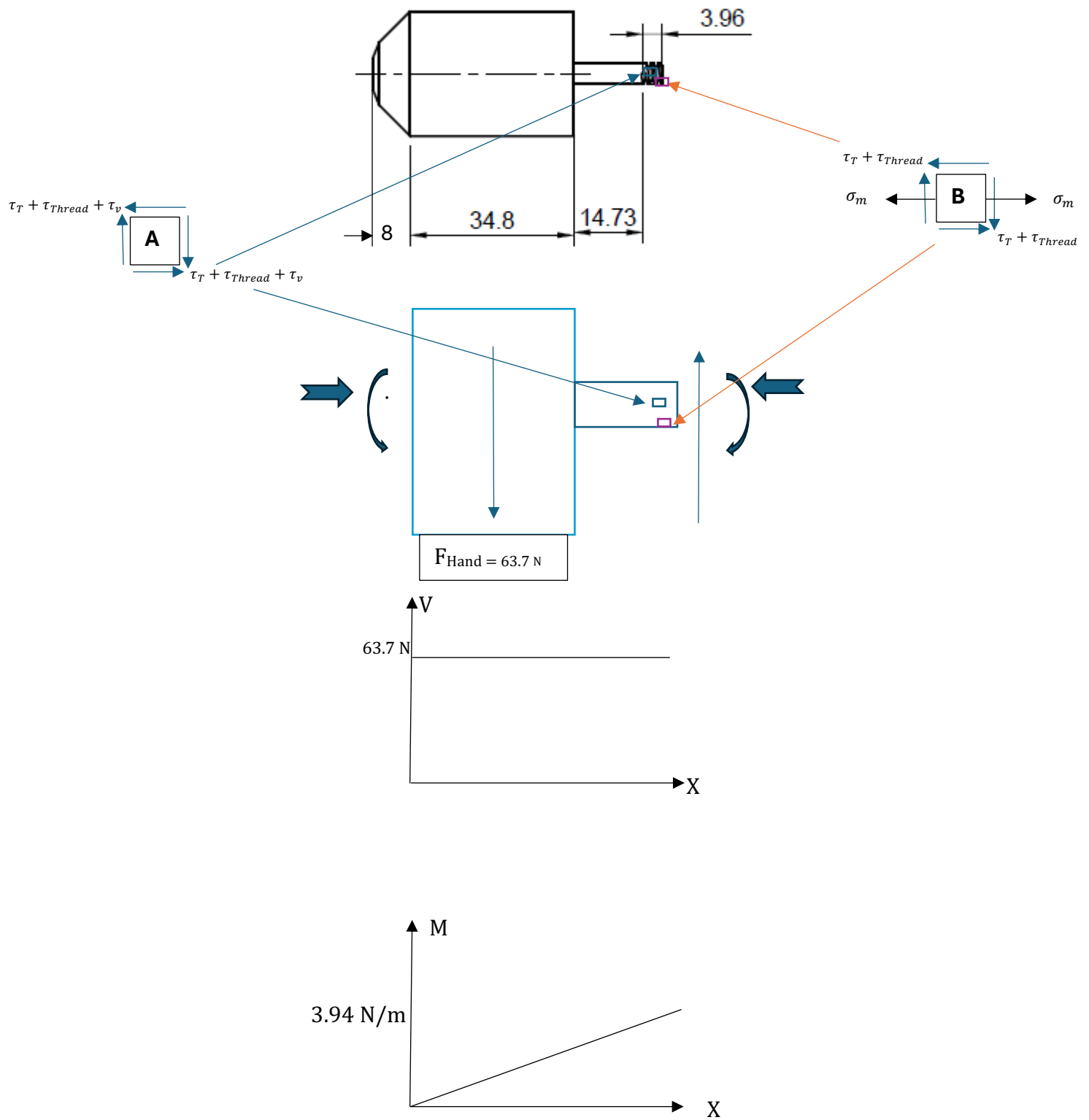
10	1	TopSpring	Steel
9	1	HelicalSpring	Steel
8	1	Springs	Steel
6	1	Slide Link	ABS
5	1	HandleLink	ABS
4	1	Trigger Release	ABS
3	1	Plunger_gasket_su bassy	ABS
1	1	Internal Frame	ABS
Item	Qty	Part Number	Material
Parts List			

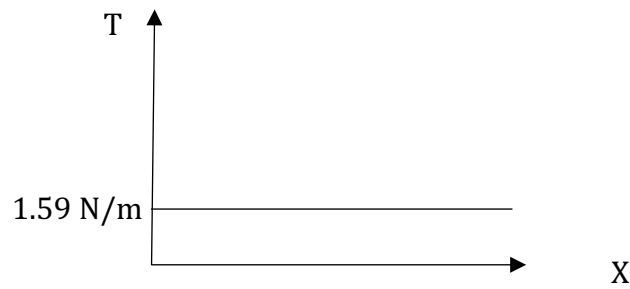
1st Weak point: Priming Bolt Shaft

3rd Angle Projection



FBD and Kinematic Calculations



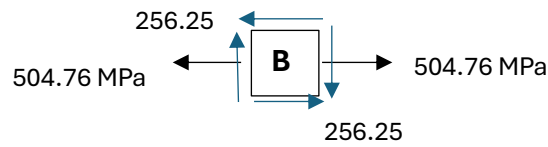


$$\tau_v = \frac{4V}{3A} = \frac{4(63.7)}{3\pi(2.15 \times 10^{-3})^2} = 5.84 \text{ MPa}$$

$$\tau_t = \frac{16T}{\pi d^3} = \frac{16(1.59)}{\pi(4.3 \times 10^{-3})^3} = 101.85 \text{ MPa}$$

$$\tau_{thread} = \frac{V}{A_s} = \frac{V}{\pi d(0.75)t} = 154.4 \text{ MPa}$$

$$\sigma_m = \frac{32M}{\pi d^3} = \frac{32(3.94)}{\pi(4.3 \times 10^{-3})^3} = 504.76 \text{ MPa}$$



Stress at Point B is maximum and is likely to fail first

$$\sigma_1, \sigma_2 = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_x = 504.76 \text{ MPa}$$

$$\sigma_y = 0$$

$$\tau_{xy} = \tau_t + \tau_{thread} = 256.25$$

On solving we get

$$\sigma_1 = 612.04 \text{ MPa}$$

$$\sigma_2 = -107.2 \text{ MPa}$$

For 4140 Alloy Steel Material, $\sigma_Y = 900$,

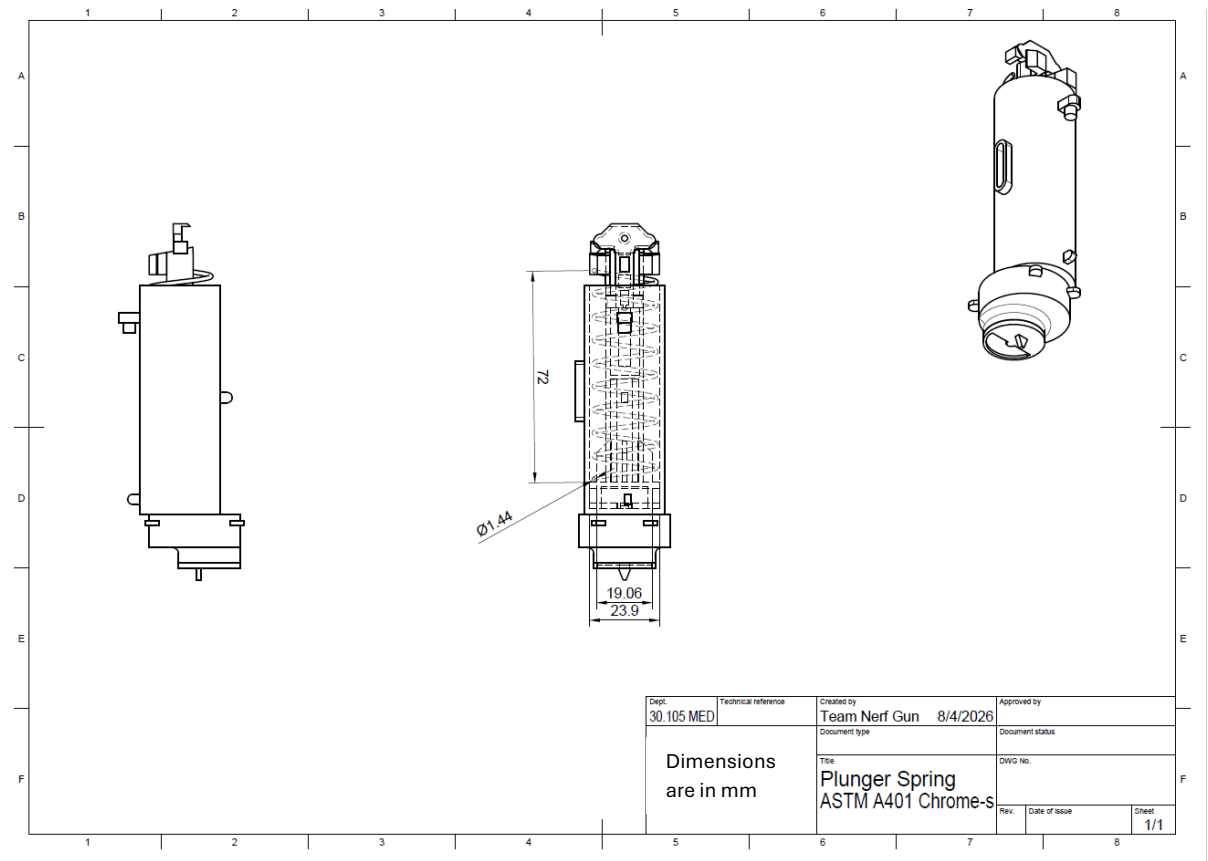
$$\text{Von Misses S.F.} = \frac{\sigma_Y}{\sqrt{\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2}} = 1.33$$

$$\tau_{abs} = \frac{\sigma_1 - \sigma_2}{2} = 360 \text{ MPa}$$

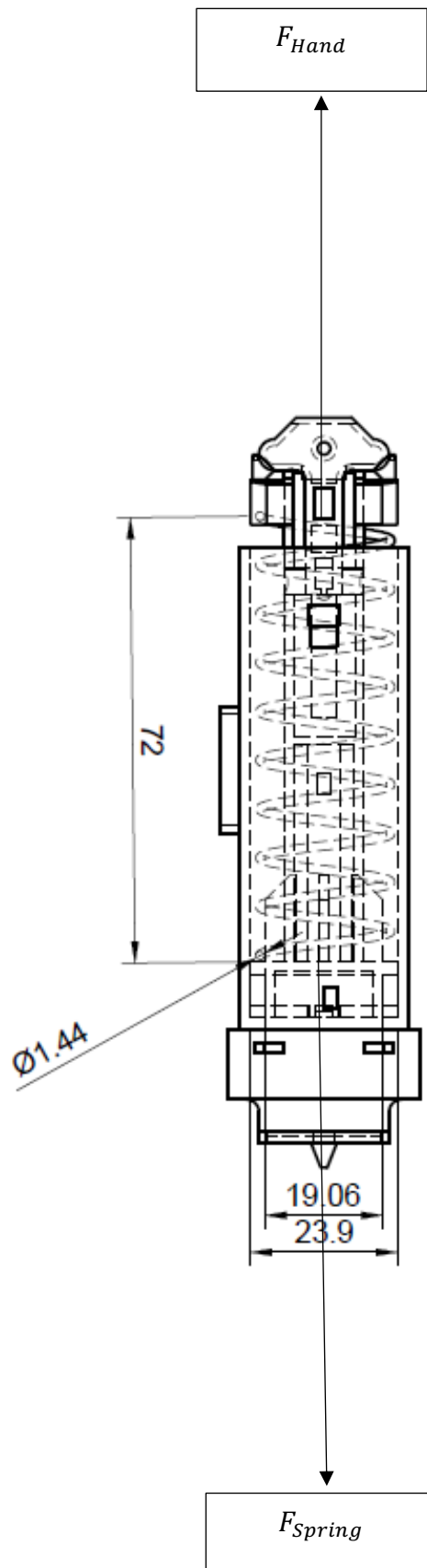
$$\text{Shear Stress Safety Factor} = \frac{0.5\sigma_Y}{\tau_{abs}} = 1.25$$

2nd Weakpoint: Plunger Spring

3rd Angle Projection



FBD & Kinematic Calculations



Free Length, $L_F = 70mm$

Outer diameter, $D_o = 22.1mm$

Inner diameter, $D_i = 19.4mm$

Mean diameter, $D = 20.75 mm$

Wire diameter, $d = 1.35mm$

Total no. of coils, $N_t = N = 9$ (Plain ends)

Pitch, $p = \frac{L_F}{N_t} = 7.7mm$

The spring is made of ASTM A401 Chrome-s,

Ultimate tensile stress, $S_U = A(d^b) = 2002MPa$

Shear modulus, $G = 78.5GPa$

$S'_n = 0.5S_U = 810MPa$

$C_L = 1$ (axial)

$C_G = 0.7$

$C_S = 0.55$ (Cold – drawn)

$C_T = 1$ (RTP)

$C_R = 0.897$ (90%)

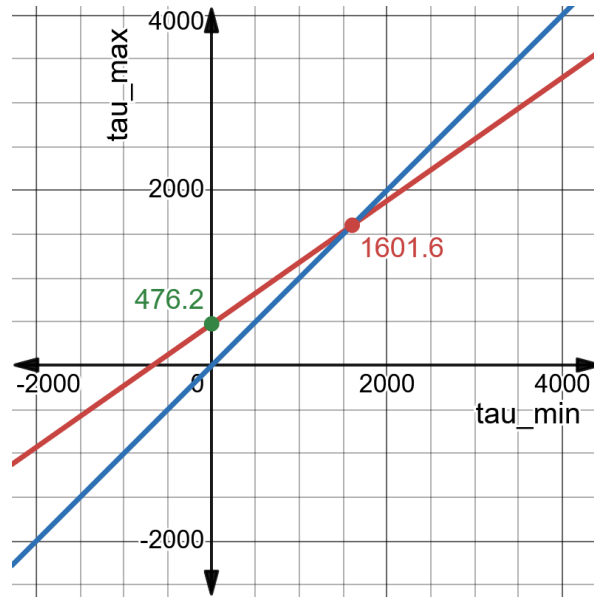
$S_n = S'_n C_L C_G C_S C_T C_R = 279.7MPa$

Ultimate shear stress, $S_{US} = 0.8 \cdot S_U = 1601.6MPa$

Minimum load, $F_{min} = 0N$

Maximum load, $F_{max} = 63.7N$

Spring constant, $k = \frac{d^4 G}{8D^3 N} = 405.38 N/m$



$\tau_{max} - \tau_{min}$ Graph

$$S_{NSum} = \frac{2S_{US}S_N}{S_{US} + S_N} = 476.2MPa$$

$$S_{max} = \frac{S_{US}S_{NSum}}{S_{US} + (S_{NSum} - S_{US})\frac{F_{min}}{F_{max}}} = S_{NSum} = 476.2MPa$$

$$\text{Spring index, } C = \frac{D}{d} = 15.4$$

$$\text{Wahl dynamic loading factor, } K_w = \frac{4C-1}{4C-4} + \frac{0.615}{C} = 1.09$$

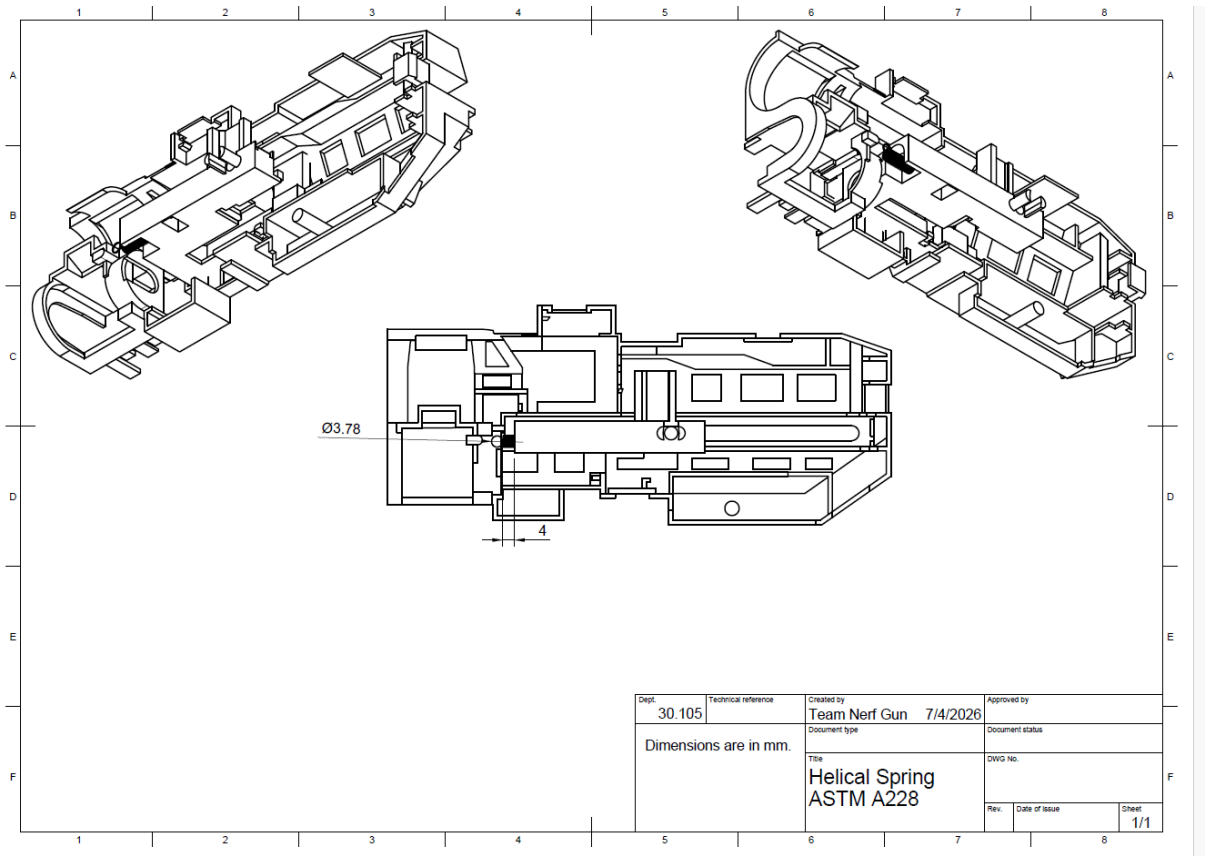
$$\text{Measured shear force applied, } F_s = 10.8N$$

$$\text{Shear spring stress, } \tau_s = \frac{8F_s}{\pi d^2} C K_w = 253.3 MPa$$

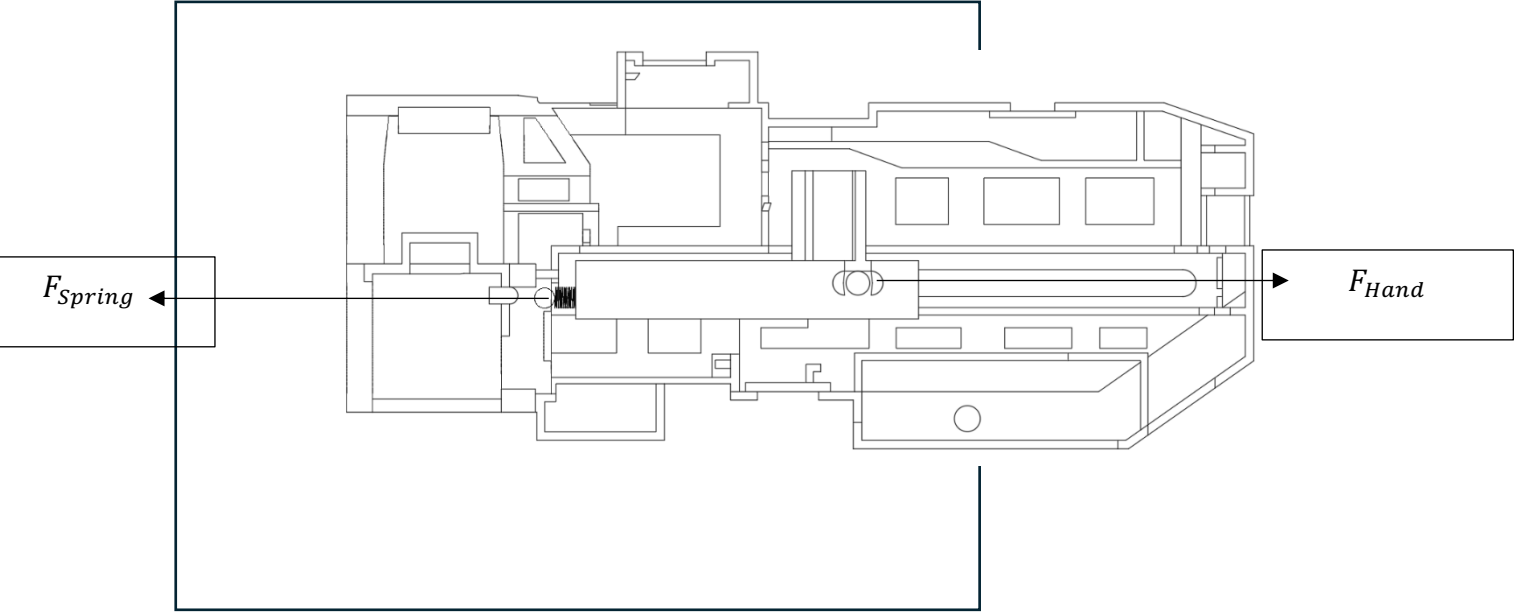
$$\text{Safety Factor, } S_F = \frac{S_{max}}{\tau_s} = 1.88$$

3rd Weak point: Extension Spring

3rd Angle Projection



FBD & Kinematic Calculations



$$r_1 = 2.1$$

$$r_3 = 2.1$$

Wire diameter, $d = 0.02\text{mm}$

Mean diameter, $D = 3\text{mm}$

No. of Coils, $N = 40$

Assuming the spring is made of ASTM A228 Music wire,

Shear modulus, $G = 79.3\text{ GPa}$

Ultimate Tensile Stress, $S_U = A(d^b) = 4066.5\text{MPa}$

$$F_{min} = 0\text{ N}$$

$$\text{Spring constant, } k = \frac{d^4 G}{8D^3 N} = 1.46 \cdot 10^{-3} \text{ N/m}$$

$$\text{Max extension, } \delta_{max} = 54\text{mm}$$

$$\text{Max load, } F_{max} = k\delta_{max} = 7.8 \cdot 10^{-5} \text{ N}$$

$$S'_n = 0.5 \cdot S_U = 793\text{MPa}$$

$$C_L = 1 \text{ (axial)}$$

$$C_G = 0.7 \text{ (} d < 10\text{mm)}$$

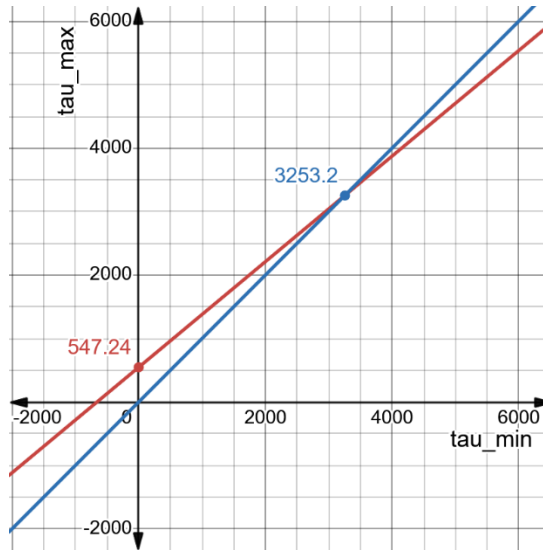
$$C_S = 0.6 \text{ (Assume cold – drawn)}$$

$$C_T = 1 \text{ (RTP)}$$

$$C_R = 0.897 \text{ (90\% Reliability)}$$

$$S_n = S'_n C_L C_G C_S C_T C_R = 298.75 \text{ MPa}$$

$$\text{Ultimate shear stress, } S_{US} = 0.8 \cdot S_U = 3253.2\text{MPa}$$



$\tau_{max} - \tau_{min}$ Graph

$$S_{NSum} = \frac{2S_{US}S_N}{S_{US} + S_N} = 547.24 \text{ MPa}$$

$$S_{max} = \frac{S_{US}S_{NSum}}{S_{US} + (S_{NSum} - S_{US})\frac{F_{min}}{F_{max}}} = S_{NSum} = 547.24 \text{ MPa}$$

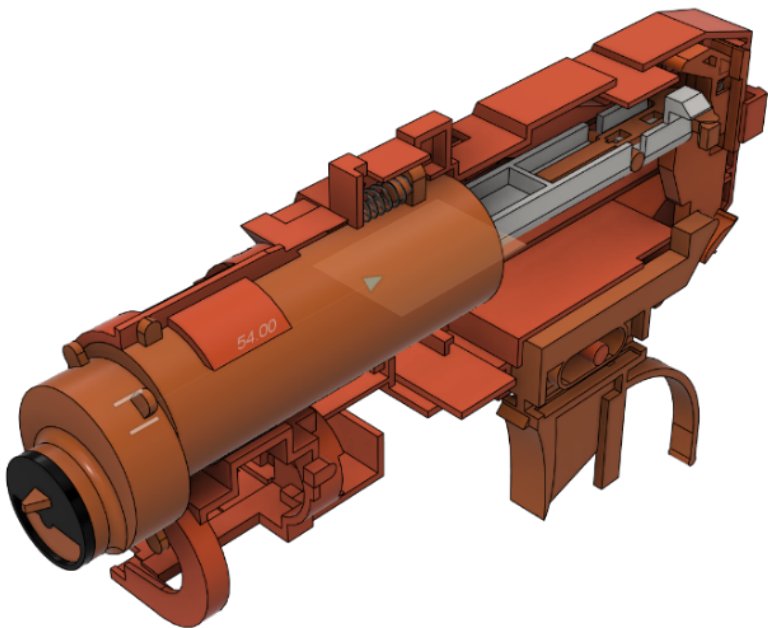
$$\text{Spring index, } C = \frac{D}{d} = 150$$

$$\text{Wahl dynamic loading factor, } K_w = \frac{4C-1}{4C-4} + \frac{0.615}{C} = 1$$

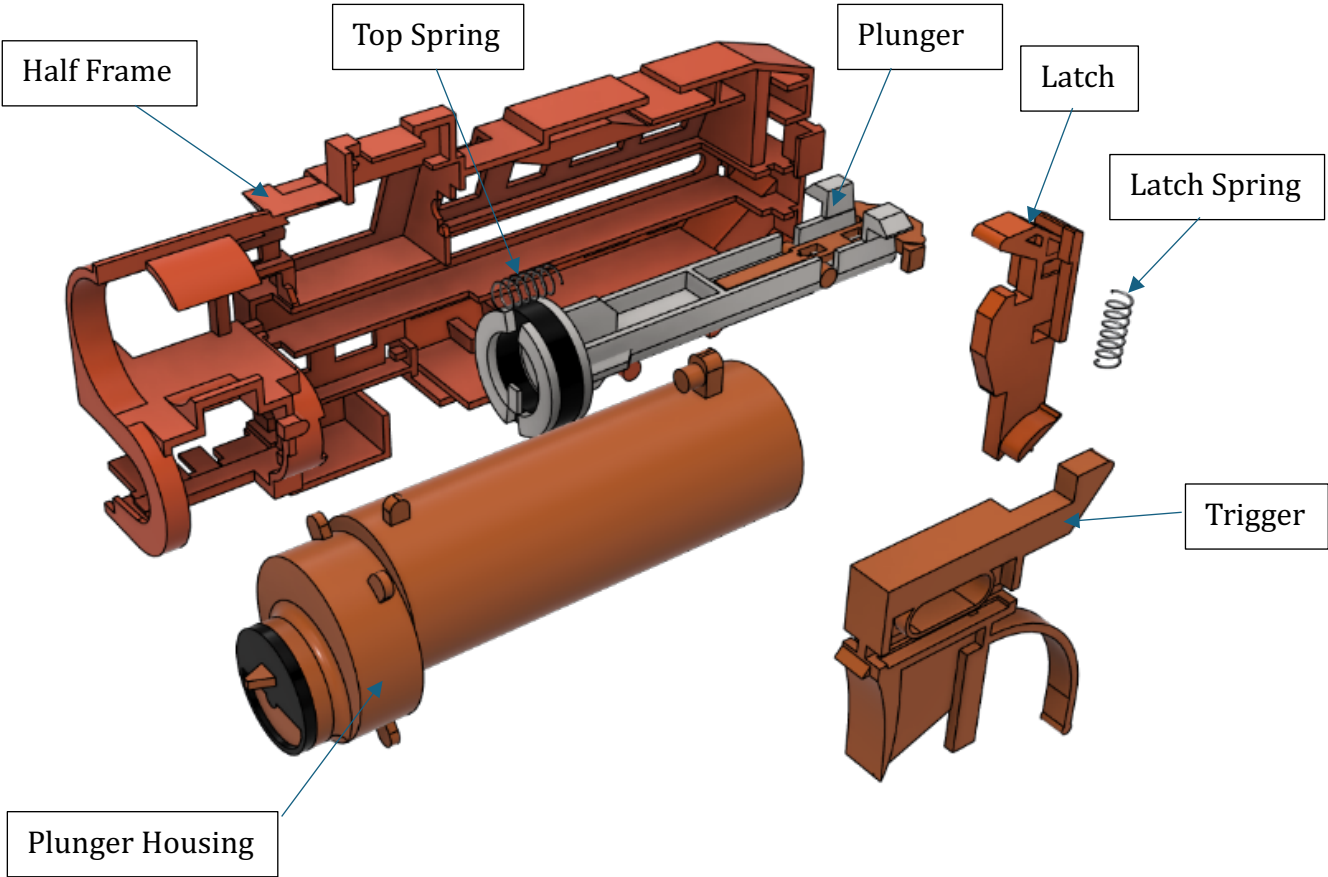
$$\text{Shear stress, } \tau_s = \frac{8F_{max}}{\pi d^2} CK_w = 74.4 \text{ MPa}$$

$$\text{Safety Factor, } S_F = \frac{S_{max}}{\tau_s} = 7.36$$

Firing Mechanism Subsystem Overview Diagram



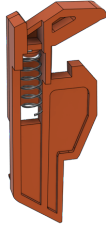
Isometric View



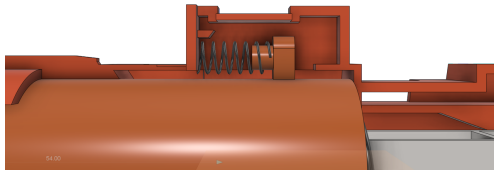
Exploded View

Firing Process:

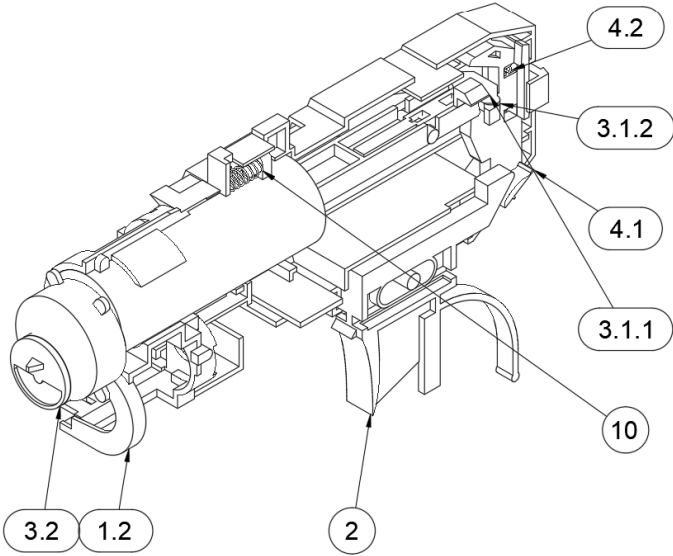
1. When the blaster is primed, the plunger is pulled back and hooked on to the latch, keeping the plunger primed, and the spring within the plunger housing compressed.
2. When the trigger is pushed back by an external force, it pushes against the latch, pushing it down and compressing the spring keeping it upright



3. As the latch is pushed down, it releases the plunger, causing it to shoot forwards in its housing, releasing the necessary force to launch the bullet in the drum.
4. This release slightly pushes the plunger housing forward, which compresses the spring between the frame and the plunger housing.



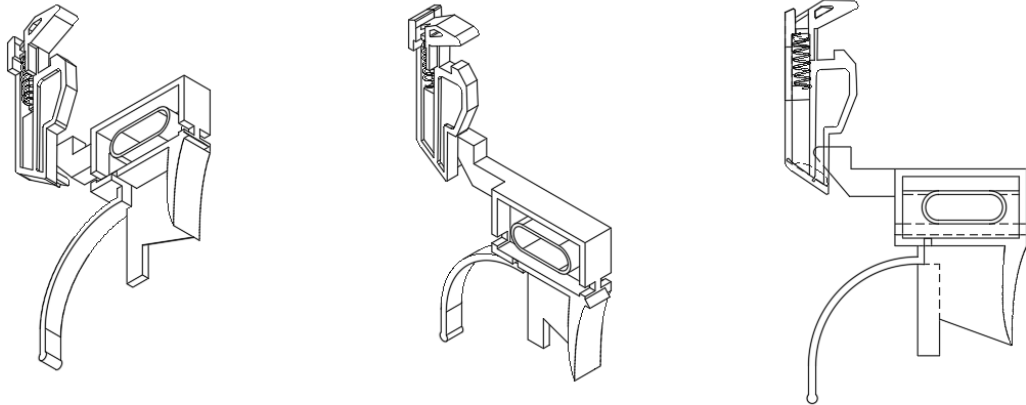
BOM Sheet (Firing Mechanism)



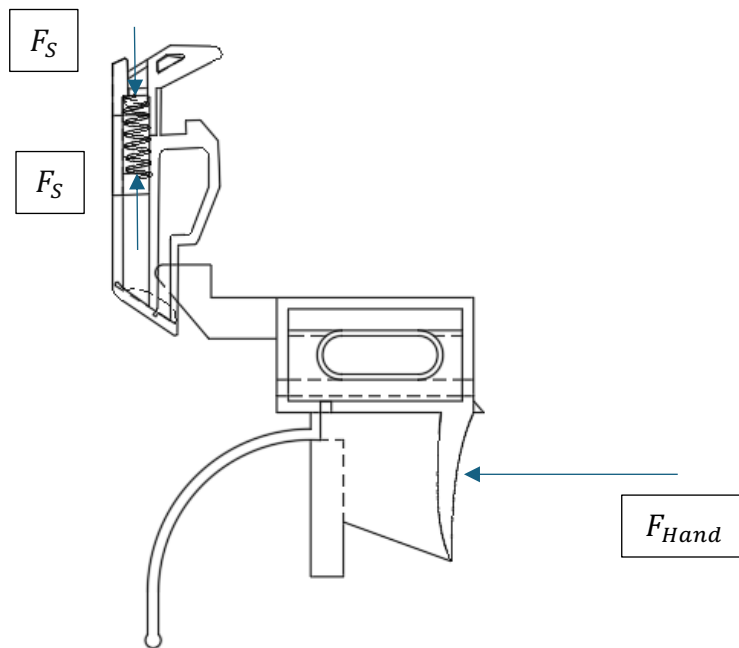
Parts List			
Item	Qty	Part Name	Material
1	1	Internal Frame v37	----
1.2	1	Internal Frame Left	ABS Plastic
2	1	Trigger v5	ABS Plastic
3	1	Plunger_gasket_subassy v16	----
3.1	1	SubAssembly_1 v2	----
3.1.1	1	Gaskethousing v8	----
3.1.2	1	Gasketattachpiece v9	ABS Plastic
3.2	1	PlungerSpringHousing v14	----
4	1	Trigger 2 v4	----
4.1	1	Latch	ABS Plastic
4.2	1	Spring	Steel
10	1	TopSpring v8	Steel

1st Weak point: Trigger Compression Spring

3rd Angle Projection



FBD & Kinematic Calculations



FBD

Assuming friction to be negligible,

Considering the spring is made from ASTM A232 wire,

Coefficient $A = 1909.9 \text{ MPa}$

Mean diameter, $D = 3.99 \text{ mm}$

Wire diameter, $d = 0.78 \text{ mm}$

Exponent $b = -0.1453$

Ultimate Tensile Stress, $S_U = A(d)^b = 1980.11 \text{ MPa}$

Measured shear force, $F_s = 4.8 \text{ N}$

Spring Index, $C = \frac{D}{d} = \frac{3.99}{0.78} = 5.11$

Wahl dynamic loading factor, $K_w = \frac{4C - 1}{4C - 4} + \frac{0.615}{C} = 1.3$

Shear stress, $\tau_s = \frac{8F_sCK_w}{\pi d^2} = 153.46 \text{ N}$

Free length, $L_f = 15.8 \text{ mm}$

Total no. coils, $N_t = N = 7.5$

Pitch, $p = \frac{L_f}{N_t} = 2.11 \text{ mm}$

$S'_n = 0.5 \cdot S_U = 990.05 \text{ MPa}$

$C_L = 1$ (axial)

$C_G = 0.7$ ($d < 10 \text{ mm}$)

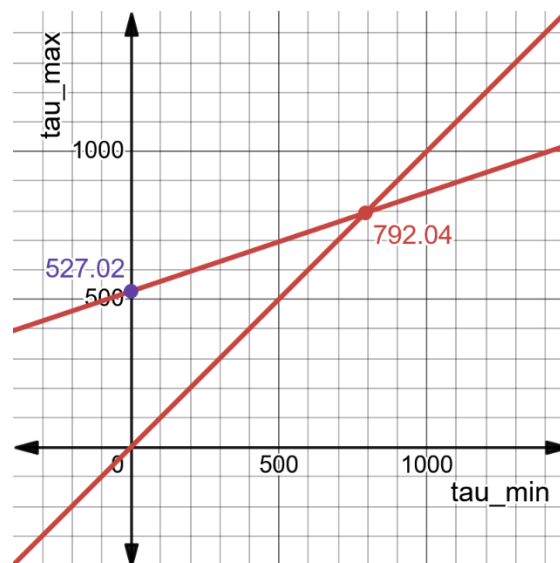
$C_S = 0.7$ (Assume cold – drawn)

$C_T = 1$ (RTP)

$C_R = 0.814$ (99% Reliability)

$S_n = S'_n C_L C_G C_S C_T C_R = 394.89 \text{ MPa}$

$S_{US} = 0.8 S_U = 792.04 \text{ MPa}$



$\tau_{max} - \tau_{min}$ Diagram

$$S_{NSum} = \frac{2S_{US}S_N}{S_{US} + S_N} = 527.02 MPa$$

Minimum load compression length, $\delta_{min} = 0.3mm$

Maximum load compression length, $\delta_{max} = 10.6mm$

$$\text{Spring constant, } k = \frac{d^4 G}{8D^3 N} = 0.12 \text{ N/m}$$

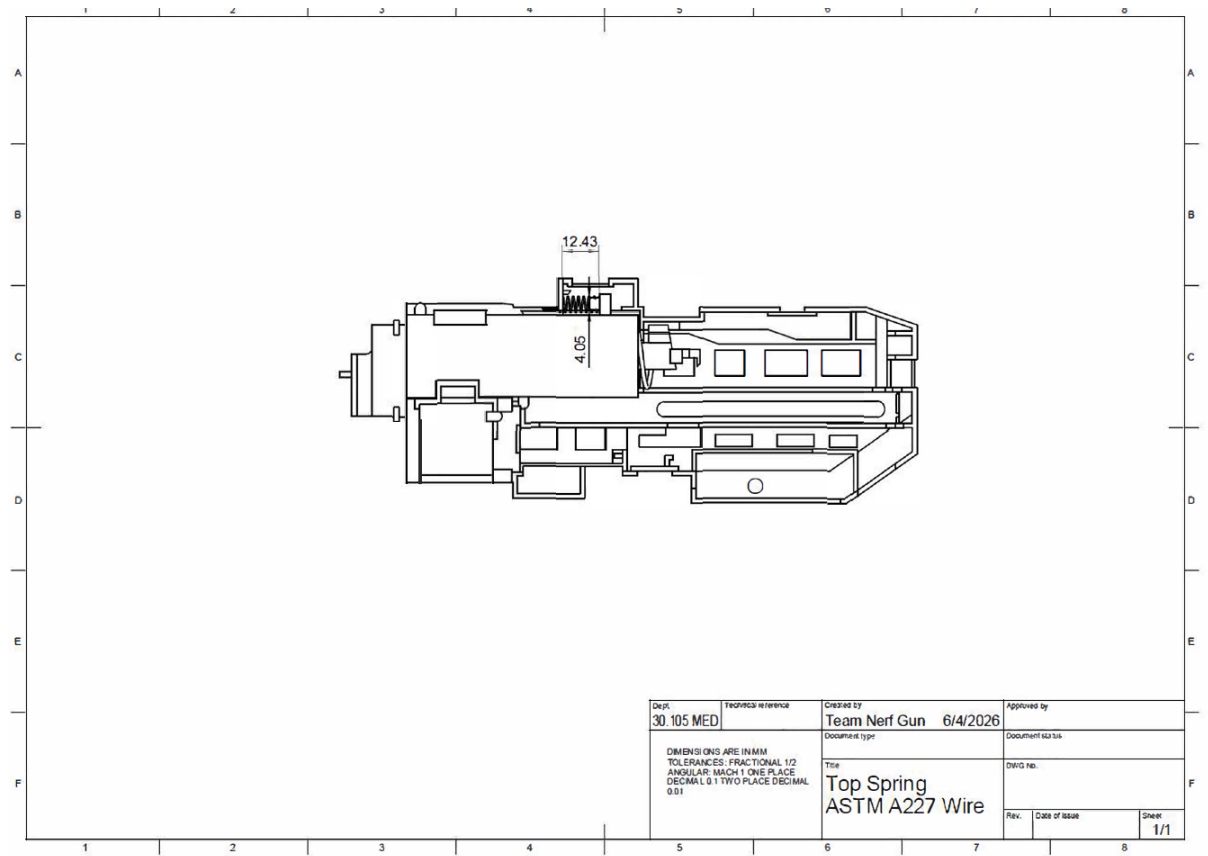
$$\frac{\tau_{max}}{\tau_{min}} = \frac{F_{max}}{F_{min}} = \frac{\delta_{max}}{\delta_i} = \frac{10.6}{0.3} = 35.3$$

$$S_{max} = \frac{S_{US} \cdot S_{NSum}}{S_{US} + (S_{NSum} - S_{US}) \frac{F_{min}}{F_{max}}} = 532.06 MPa$$

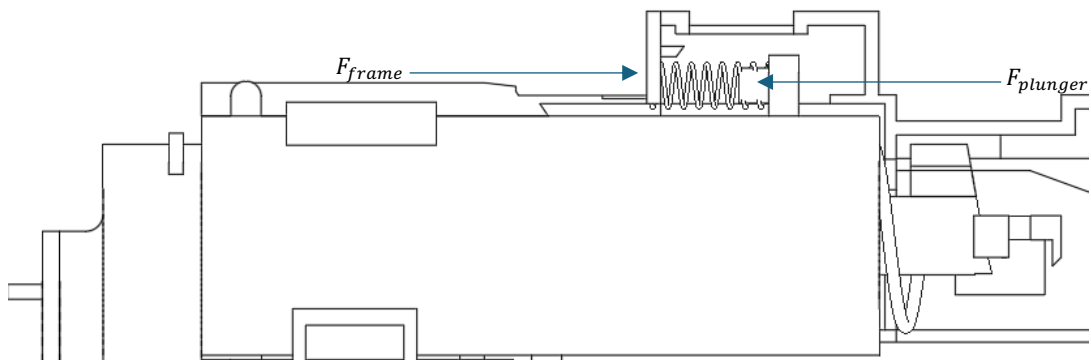
$$\text{Safety Factor, } S_F = \frac{S_{max}}{\tau_s} = 3.47$$

2nd Weak point: Top Spring (Fatigue Loading)

3rd Angle Projection



FBD & Kinematic Calculations



$$L_f = 18mm \quad D_o = 4.9mm \quad D_i = 4.15mm \quad D = 4.5mm \quad d = 0.4mm \quad N_t = N = 10 \quad m$$

$$G = 80GPa$$

The Top Spring undergoes a fatigue compressive loading with an initial preload.
Attached onto the plunger housing to absorb the shock from the plunger when it fires.

$$k = \frac{d^4 G}{8D^3 N} = 280.9 \text{ N/m}$$

Considering the spring is made from ASTM A227 wire,

Coefficient A = 1753.3 MPa

Mean diameter, D = 4.5mm

Wire diameter, d = 0.4mm

Exponent b = -0.1822

$$\text{Ultimate Tensile Stress, } S_u = A(d)^b = 2071.6 \text{ MPa}$$

$$\text{Spring Index, } C = \frac{D}{d} = \frac{4.5}{0.4} = 11.25$$

$$\text{Wahl dynamic loading factor, } K_w = \frac{4C - 1}{4C - 4} + \frac{0.615}{C} = 1.12$$

Spring Material = ASTM A22 Wire

$$S'_n = 0.5S_u = 0.5(2071.6) = 1035.8 \text{ MPa}$$

$$C_L = 1 \text{ (axial)}$$

$$C_G = 0.7$$

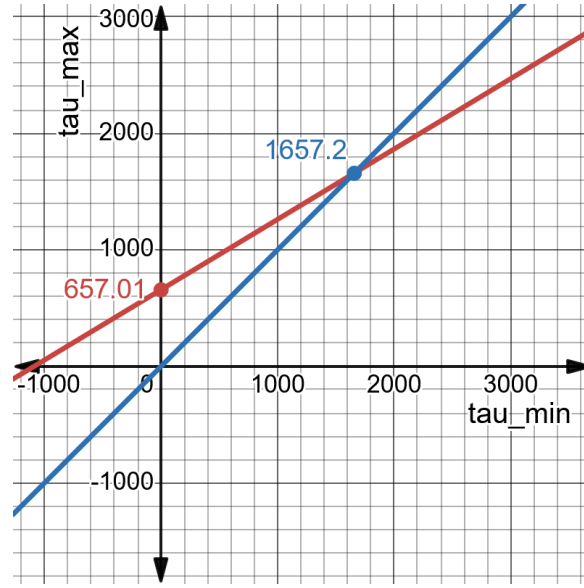
$$C_s = 0.63 \text{ (cold drawn / machined)}$$

$$C_T = 1$$

$$C_R = 0.897$$

$$S_n = S'_n C_L C_G C_s C_T C_R = 409.73 \text{ MPa}$$

$$S_{US} = 0.8S_u = 0.8(2071.6) = 1657.2 \text{ MPa}$$



$\tau_{max} - \tau_{min}$ Diagram

$$S_{NSum} = \frac{2S_{US}S_N}{S_{US}+S_N} = 657.01 \text{ MPa}$$

$$L_{initial} = 12.43 \approx 13\text{mm} \implies \delta_{initial} = 18\text{mm} - 13\text{mm} = 5\text{mm}$$

$$L_{final} = 9\text{mm} \implies \delta_{final} = 18\text{mm} - 9\text{mm} = 9\text{mm}$$

$$F_{initial} = k\delta_{initial} = 1.4\text{N}$$

$$F_{final} = k\delta_{final} = 2.5\text{N}$$

$$\delta_{Wmax} = \delta_{final} - \delta_{initial} = 4\text{mm}$$

$$\delta_{clash} = 0.1\delta_{max} = 0.9\text{mm}$$

$$\delta_s = \delta_{clash} + \delta_{Wmax} + \delta_{initial} = 9.9\text{mm}$$

$$F_s = k\delta_s = 2.78\text{N}$$

$$\frac{\tau_{max}}{\tau_{min}} = \frac{F_{max}}{F_{min}} = \frac{\delta_{max}}{\delta_i} = \frac{9}{5} = 1.8$$

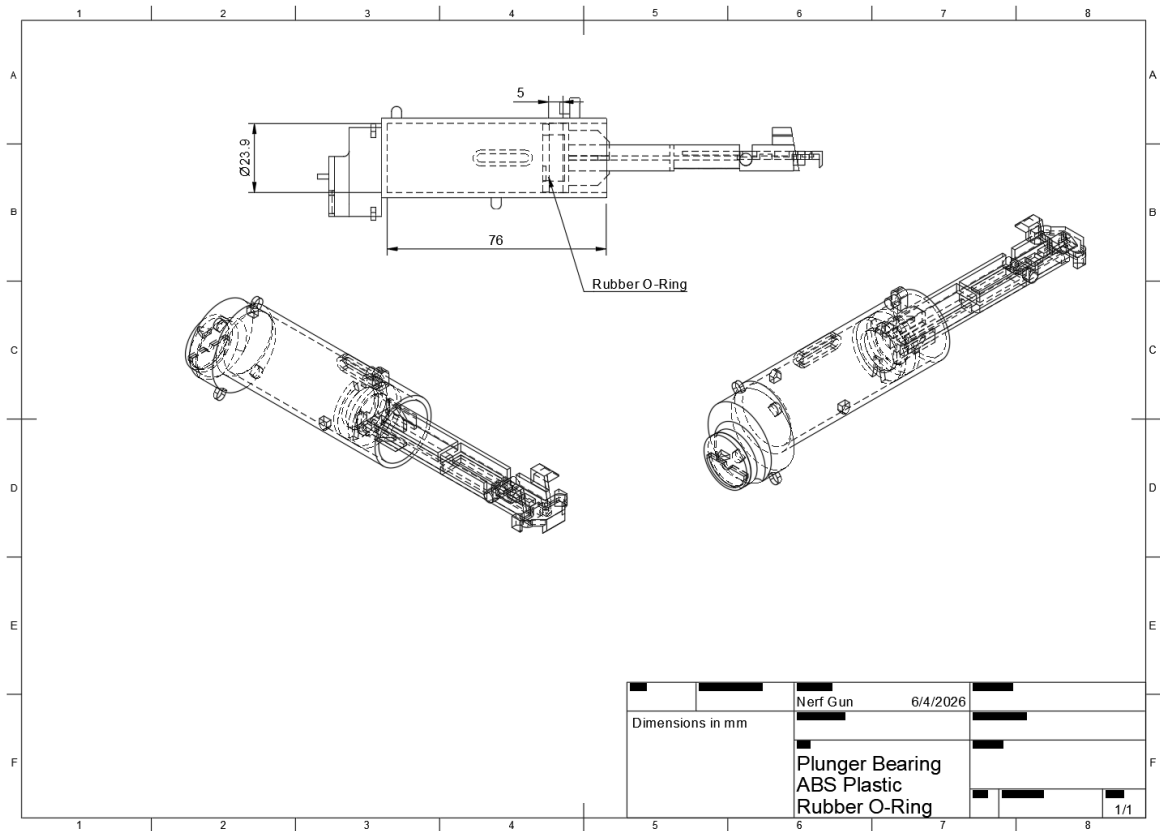
$$\text{Shear stress, } \tau_s = \frac{8F_sCK_w}{\pi d^2} = 557.48\text{N}$$

$$S_{max} = \frac{S_{US} \cdot S_{NSum}}{S_{US} + (S_{NSum} - S_{US}) \frac{F_{min}}{F_{max}}} = 987.28 \text{ MPa}$$

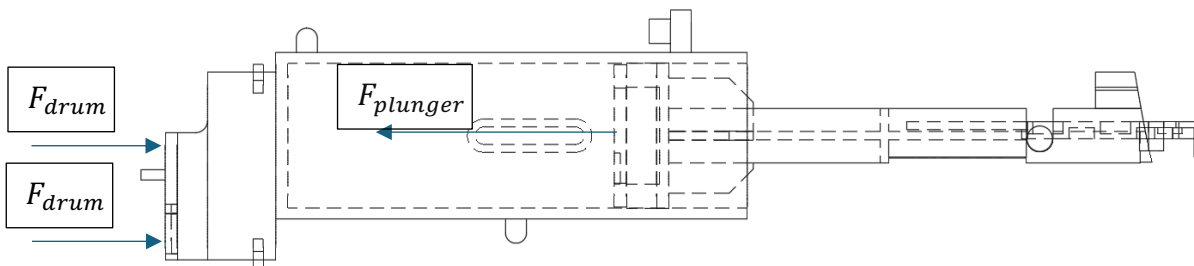
$$\text{Safety Factor, } S_F = \frac{S_{max}}{\tau_s} = 1.77$$

3rd Weakpoint: Plunger Plain Bearing

3rd Angle Projection



FBD & Kinematic Calculations



FBD

Outer diameter, $D_o = 27.5\text{mm}$

Inner diameter, $D_i = 24.1\text{mm}$

Length, $L = 78\text{mm}$

Plunger diameter, $d = 24\text{mm}$

$F_{bearing} = F_{plunger} = 63.7\text{N}$

Bearing pressure, $P = \frac{F_{bearing}}{Ld} = 0.034\text{MPa}$

Mass of plunger, $m = 18.3\text{g}$

Acceleration, $a = \frac{F}{m} = 3538.8\text{m/s}^2$

Using $v^2 - u^2 = 2as$,

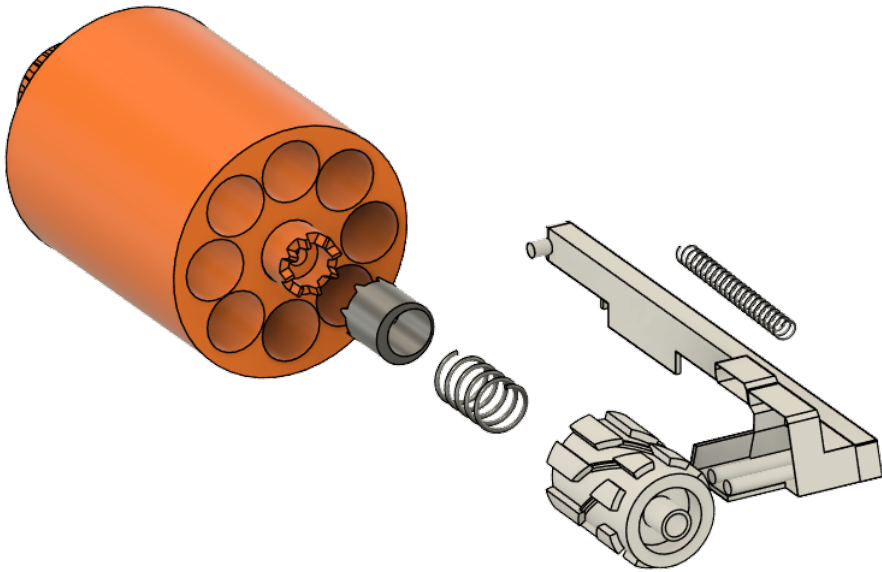
$u = 0$, $s = 55\text{mm}$

Velocity of plunger, $v = \sqrt{2as} = 19.72\text{ m/s}$

$PV = 0.67\text{ MPa} \cdot \text{m/s}$

Safety factor, $S_F = \frac{PV_{limit}}{PV} = 1.31$

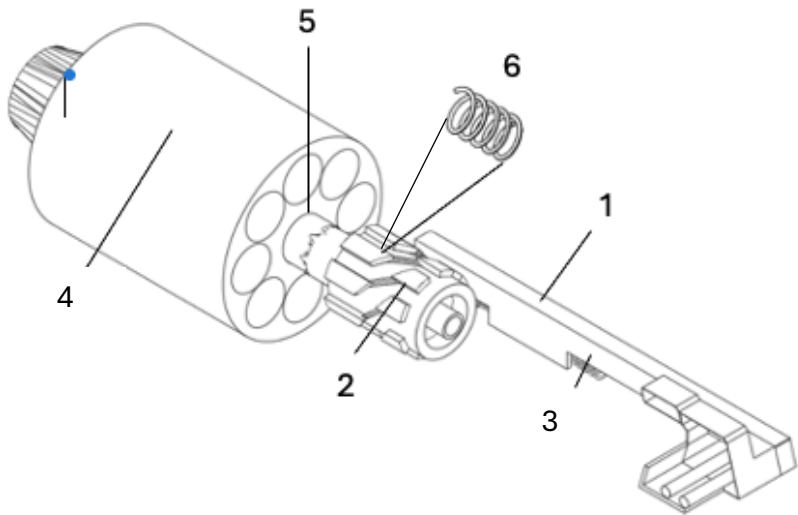
CAM Mechanism Subsystem Overview Diagram



Process:

- 1. Cam rail is pulled back, moving the flat-faced follower along the surface of the cam wheel and compressing attached spring
- 2. Cam wheel and attached hirth joint turns anti-clockwise with the movement of the follower
- 3. Hirth joint transfers motion to the dart barrel, which also moves anti clockwise.
- 4. Cam rail is released and spring returns to natural length.

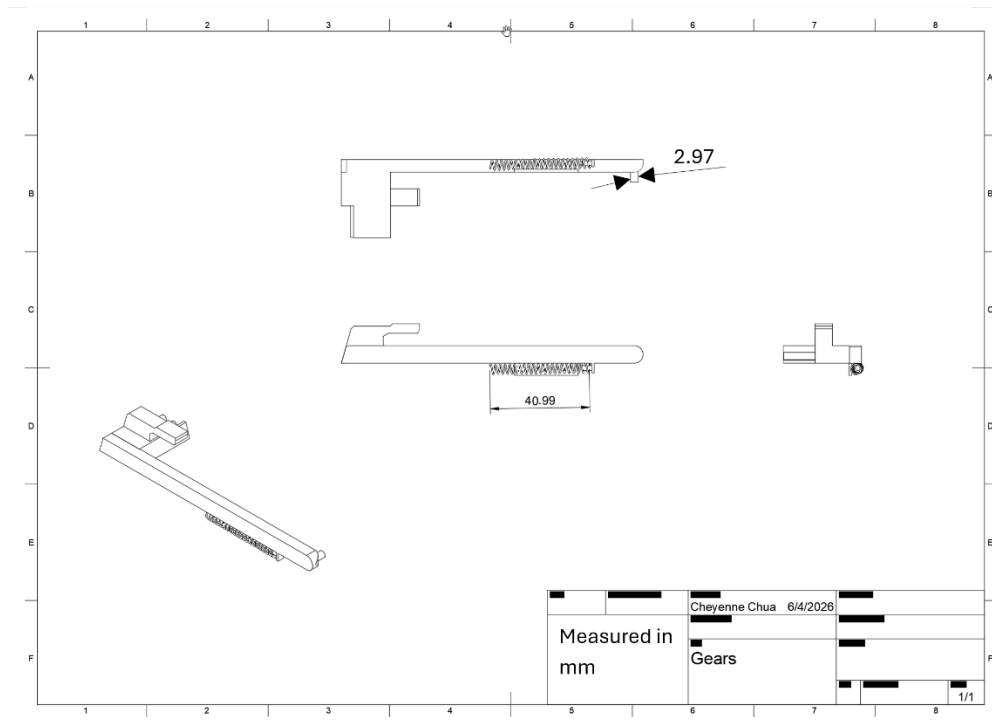
BOM Sheet (Rotating Drum)



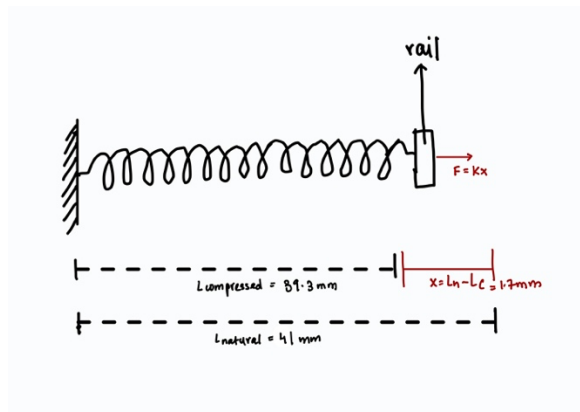
S.No	Part	Function
1.	Cam Follower	Linear Input
2.	Cam Wheel	Converts linear to rotational motion
3.	Compression Rail Spring	Restores cam follower to original position
4.	Dart Barrel	Rotates and holds darts
5.	Hirth Joint	Transmits cam rotation to barrel
6.	Inner Compression Spring	Keeps hirth joint and cam wheel in place.

Weak Point 1: Cam Follower (Rail) Spring

3rd Angle Projection



FBD & Kinematic Calculations



$$d = 0.46 \times 10^{-3}, D = 4.4 \times 10^{-3}, G \approx 79 \times 10^9, n = 21$$

$$k = \frac{d^4 G}{8 D^3 n}$$

$$k = \frac{(0.46 \times 10^{-3})^4 (79 \times 10^9)}{8 (4.4 \times 10^{-3})^3 (21)} = 247.1$$

$$x = 0.017$$

$$F = kx$$

$$F = 247.1 (0.017) = 4.201\text{N}$$

$$C = \frac{D}{d}$$

$$C = \frac{4.4}{0.46} = 9.56$$

$$K_w = \frac{4C-1}{4C-4} + \frac{0.615}{C}$$

$$K_w = \frac{4(9.56)-1}{4(9.56)-4} + \frac{0.615}{9.56} = 1.1519$$

$$F_{max} = 4.201\text{N (compressed)}$$

$$F_{min} = 0 \text{ (natural length, no preload)}$$

$$\tau_{max} = 556.8$$

$$\tau_{min} = 0$$

Spring is steel, $S_u \approx 1500\text{MPa}$

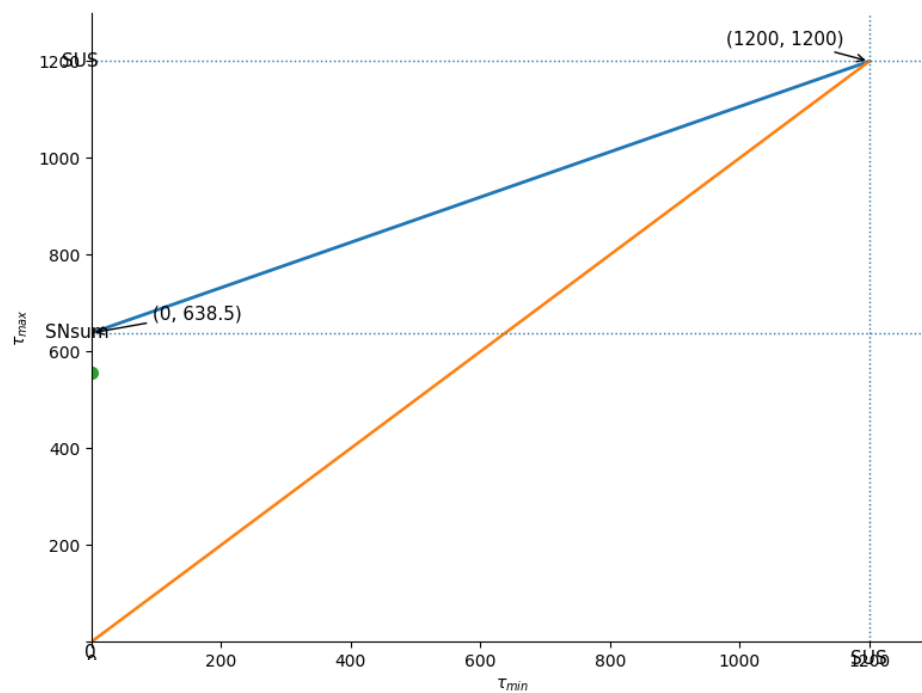
$$S_{us} = 0.8S_u = 1200 \text{ MPa}$$

$$S_N = 0.29(1500) = 435\text{MPa}$$

$$S_{Nsum} = \frac{2S_{us}S_N}{S_{us}+S_N} = 638.5 \text{ MPa}$$

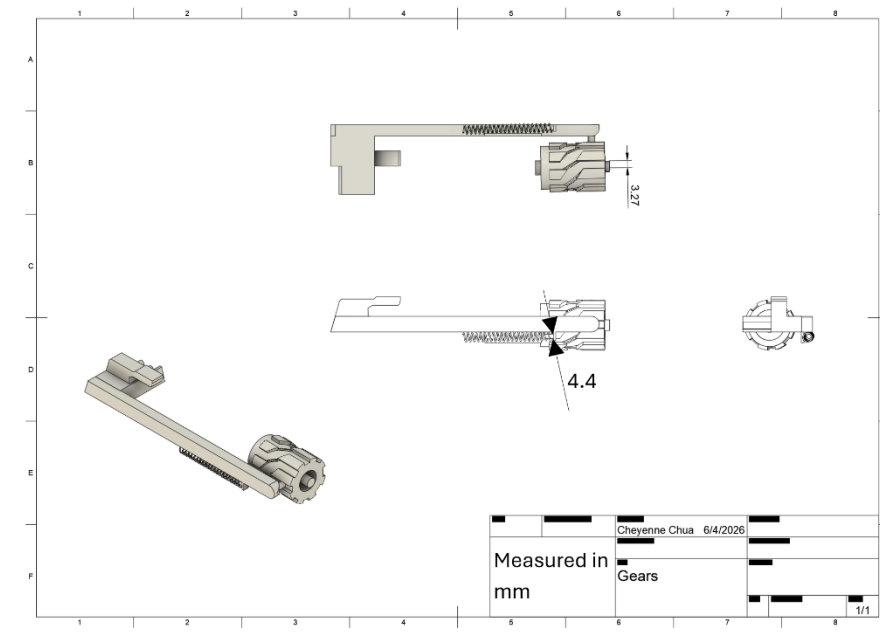
$$S_{max} = \frac{S_{us}S_{Nsum}}{S_{us}+(S_{Nsum}-S_{us})\frac{\tau_{min}}{\tau_{max}}} = 638.5 \text{ MPa}$$

$$S_F = \frac{638.7}{556.8} = 1.15$$

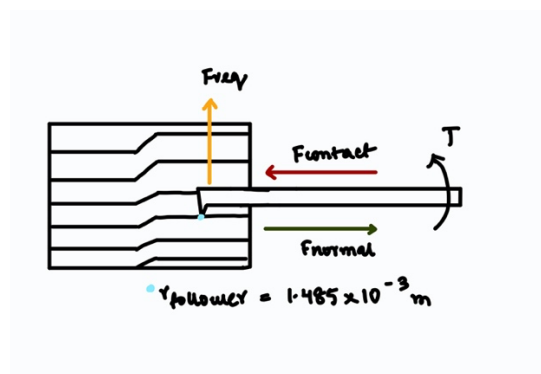


2nd Weak point: Cam Follower

3rd Angle Projection



FBD & Kinematic Calculations



Mass of follower: $m = 7.5 \times 10^{-3} \text{ kg}$

$F_{\text{available}} = \text{spring}$

$F_{\text{contact}} = 4.2$

$\mu = 0.3, F_{\text{normal}} \approx 1.5 \text{ N (spring + weight)}, r_{\text{cam}} = 0.004 \text{ m}$

$T_{\text{friction}} = \mu F_{\text{normal}} r_{\text{barrel}}$

$T_{\text{friction}} = (0.3)(1.5)(0.004) = 1.8 \times 10^{-3} \text{ N}$

$$\underline{r_{follower} = 1.485 \times 10^{-3} \text{m}}$$

$$F_{req} = \frac{T_{friction}}{r_{follower}}$$

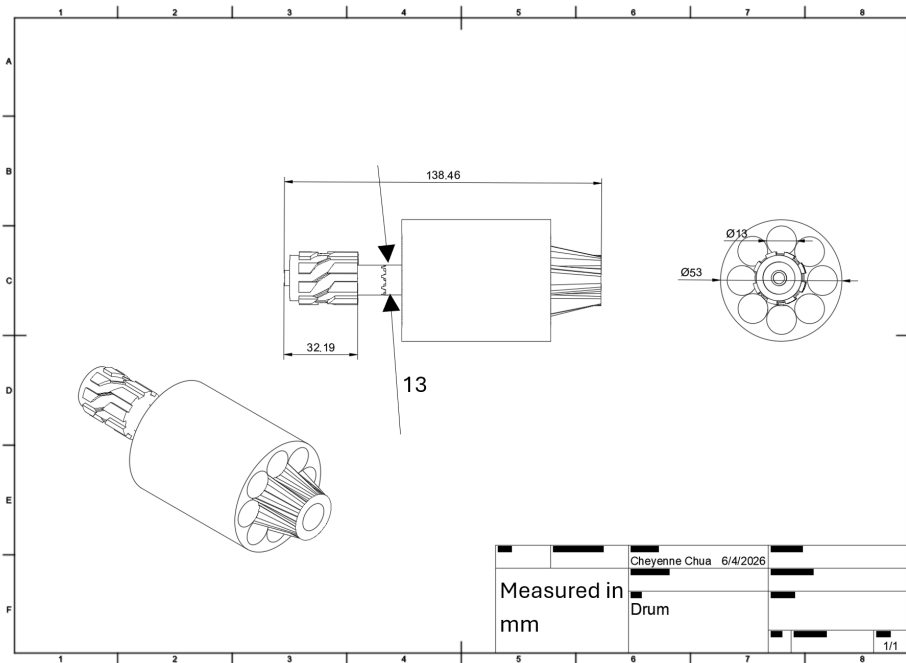
$$F_{req} = \frac{1.8 \times 10^{-3}}{1.485 \times 10^{-3}} = 1.21 \text{N}$$

$$S_F = \frac{F_{Spring}}{F_{req}}$$

$$S_F = \frac{4.2}{1.21} = 3.47$$

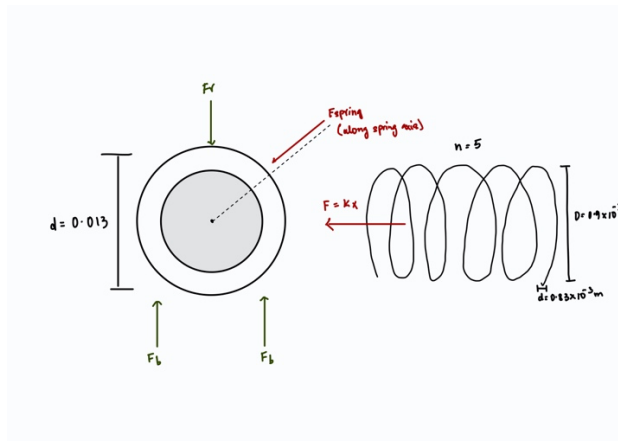
3rd Weak point: Barrel and Shaft

3rd Angle Projection



FBD & Kinematic Calculations

Measured in mm



For the inner spring,

$$\underline{d = 0.83 \times 10^{-3}, D = 0.90 \times 10^{-3}, G \approx 79 \times 10^9, n = 5}$$

$$k = \frac{d^4 G}{8 D^3 n}$$

$$k = \frac{(0.83 \times 10^{-3})^4 (79 \times 10^9)}{8 (0.9 \times 10^{-3})^3 (5)} = 1285736 \text{ N/m}$$

$$\underline{x = 1.4 \times 10^{-3}}$$

$$F_{spring} = kx$$

$$F_{spring} = 1285736(1.4 \times 10^{-3}) = 1800\text{N}$$

$$F_{spring} = F_t, \theta \approx 20^\circ$$

$$F_r = F_{spring} \tan \theta$$

$$F_r = 1800 \tan 20^\circ = 655\text{N}$$

$$F_{bearing} \approx \frac{F_r}{2}$$

$$F_{bearing} \approx \frac{655}{2} \approx 327.5\text{N}$$

$$d_{bearing} = 0.013$$

$$\underline{L = 1.9 \times 10^{-3}}$$

$$P = \frac{F_{bearing}}{dL}$$

$$P = \frac{327.5}{(0.013)(1.9 \times 10^{-3})} = 13.259\text{MPa}$$

Assuming allowable pressure of ABS: $P_{allowable} \approx 15\text{MPa}$

$$S_F = \frac{P_{allowable}}{P}$$

$$S_F = \frac{15}{13.259} = 1.13$$

Contributions

Rey: Priming & firing mechanisms

Ojas: Priming & firing mechanisms

Cheyenne: Rotating barrel drum mechanism

Raina: Rotating barrel drum mechanism